GLOBAL APPLIED KNOWLEDGE

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Week - 6 Pandas

Pandas :-

It is a python library used for :

- Reading and analysing data.
- working with csv and excel files.
- Organising data into tables
- filtering and sorting data
- calculating totals, averages and percentages
- Grouping and summarising data.

<!> python

→ Import pandas as pd.
pd is the commonly used short name for pandas.

Series and Dataframe.

- Series :

  A series is a one-dimensional data structure. It represents a single column of data with an index.

Important series functions :

Functions	purpose
pd.series( )	Creates a series
• max ( )	finds maximum value
• min ( )	finds minimum valu
• sum ( )	calculate total
• mean ( )	calculate avg.

- Dataframe :

  A dataframe is a two-dimentional data structure containing rows and columns, like an excel spreadsheet.

- Important Dataframe properties.

```
print(df.head())
print(df.tail())
print(df.shape)
print(df.dtypes)
print(df.colums
print(len(df))
```

Selecting and filtering Data

- Select rows using iloc and loc
  There are two ways to select rows
  iloc - Integer location
  ⇒ selects rows using their numerical position.
  loc - label location
  ⇒ select rows using their index labels.

iloc

Uses position

df.iloc[0]

position starts at 0

loc

uses index label

df.loc[0]

uses actual index labels

- Combining conditions

AND (&)

Both conditions must be true

OR (|)

Atleast one condition must be true.

Pandas - Notes >> continue.

Grouping, Sorting and Aggregating

A • groupby()

groupby() divides data into groups based on a particular column.

B • Important aggregation functions

Function	purpose
• mean()	Average
• sum()	Total
• max()	Maximum
• min()	Minimum
• count()	Number of non-missing values
• size()	Number of rows in each group

C. .agg()

use .agg() when you want multiple calculations together.

D. value_counts()

counts how many times each value appears in a column.

E. Sorting data

use sort_values() to arrange rows.

→ Ascending order (smallest to largest)

<1> python

print(df.sort_values("percent"))

→ Descending order (largest to smallest)

<1> python

print(df.sort_values("percent", ascending = False))

.apply . lambda and Finishing up

A. . apply ()

The . apply () method applies a function to every value in a series or along an axis of a .Data frame.

B. lambda function

ex lambda is a small anonymous function that can be written in one line.

C. Conditional lambda

ex lambda can also use conditional expressions.

D. . apply () with lambda

This is one of the most important combinations

E.g. • describe()
provides statistical information about numerical columns..

Important functions:

Function	purpose
plt.title()	Adds chart title
plt.xlabel()	Adds X-axis label
plt.ylabel()	Adds y-axis label
plt.savefig()	saves chart
plt.show()	Displays chart